



Erlang's loss system and B -formula

Intro...

- Erlang's B-formula is based on the model, described by the three elements **structure, strategy, and traffic**.
 - **Structure**: We consider a system of n identical channels (servers, trunks, slots) working in parallel.
- This is called a homogeneous group.
 - **Strategy**: A call arriving at the system is accepted for service if at least one channel is idle.
 - Every call needs one and only one channel. We say the group has full accessibility.

Cont...

- If all channels are busy the system is congested and call attempts are blocked.
- A blocked (= rejected, lost) call attempt disappears without any after-effect as it may be accepted by an alternative route.
- This is called Erlang's loss model or the Blocked Calls Cleared = BCC{model

- Within a full accessible group we may look for an idle channel in different ways:
- **Random hunting**: we choose a random channel among the idle channels.
- On average every channel will carry the same traffic.
- **Ordered hunting**: the channels are numbered $1; 2; \dots; n$; and we search for an idle channel in this order, always starting with channel one (ordered hunting with homing).
- This is also called sequential hunting. A channel will on the average
- carry more traffic than the following channels.
- **Cyclic hunting**: this is similar to ordered hunting, but without homing.
- We continue hunting for an idle channel starting from the position where we ended last time. Also in this case every channel will on the average carry the same traffic.

- **Traffic:** In the following we assume that:
- The arrival process is a Poisson process with rate λ ,
- The service times are exponentially distributed with intensity μ (corresponding to a mean value $1/\mu$).
- This type of traffic is called Random Traffic (RT) or Pure Chance Traffic type One (PCT-I).

Erlang B Formula

- The Erlang B formula is expressed as GoS or probability of finding *N channels* busy.
- The assumptions in the Erlang B formula are:
 - Traffic originates from an infinite number of traffic sources independently.
 - Lost calls are cleared (BCC) assuming a zero holding time.

Cont...

- Number of trunks or service channels is limited.
- Inter-arrival times of call requests are independent of each other.
- The probability of a user occupying a channel (called service time) is based on an exponential distribution.
- Traffic requests are represented by a Poisson distribution implying exponentially distributed call inter-arrival times.

Cont...

- Erlang B Formula provides an estimate of a call blocking probability used as a measure of grade of service in telephone systems when a mean traffic load of A Erlangs is applied to a trunked system of N channels.

$$G_B = B(N, A) = \frac{A^N/A!}{\sum_{i=0}^N (A^i/i!)}$$

where:

N = number of serving channels

A = offered load, and

B(N, A) = blocking probability

Poisson's Formula

- The Poisson formula is used for designing trunks on a route for a given GoS.

The assumptions in Poisson's formula are:

- Traffic originates from an infinite number of independent sources
- Traffic density per traffic source is equal
- Lost calls are held (BCH)
- A limited number of trunks or service channels exist


$$p_b = e^{-A} \sum_{i=N}^{\infty} \frac{A^i}{i!}$$

where:

p_b = probability of blocking

A = offered load, and

N = number of trunks or service channels

Erlang C Formula

- The Erlang C formula assumes that a queue is formed to hold all requested calls that cannot be served immediately.
- Customers who find all N servers *busy* join a queue and wait as long as necessary to receive service.
- This means that the blocked customers are delayed. No server remains idle if a customer is waiting.

Cont...

The assumptions in the Erlang C formula are:

- Traffic originates from an infinite number of traffic sources independently.
- Lost calls are delayed (BCD).
- Number of trunks or service channels is limited.
- The probability of a user occupying a channel (called *service time*) is based on an exponential distribution.
- Calls are served in the order of arrival.


$$C(N, A) = \frac{A^N / [N!(1 - A/N)]}{\sum_{i=0}^{N-1} \frac{A^i}{i!} + \frac{A^N}{N!(1 - A/N)}}$$

where:

N = number of service channels

A = offered load, and

$C(N, A)$ = blocking probability

Example

- How many mobile subscribers can be supported with 50 service channels at 2% GoS? Assume the average call holding time equals 120 seconds and the average busy hour call per subscriber is 1.2 calls per hour.
- Solution: In class

Example.

- If there are 60 radio channels in a cell to handle all the calls and the average call holding time is 120 seconds, how many calls are handled in this cell with a GoS of 2%?
 - a. BCC
 - b. BCD
 - c. BCH

Solution: In class